Logarithmic Profile Jets and Regular Nodal Cells

A scoped analysis of sharp nodal length on smooth surfaces

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Abstract

The assigned manuscript claims the sharp wavelength-order upper bound for nodal length on every fixed smooth closed Riemannian surface. Its proof uses subharmonic logarithmic profiles, affine corrections, stopped descent, and packing of exterior persistence witnesses. We isolate a quantitative feature of its profile subdivision: for one logarithmic atom the prescribed gradient correction has an exact disk-growth error, and its grid average really has the logarithmic loss present in the source estimate. Subtracting harmonic Taylor polynomials of any fixed degree at least two removes that loss at the profile level. We prove the accompanying coefficient costs, retain near singularities without Taylor expansion, and specify why the result is not a replacement for affine persistence. Original Ram jet identities also yield conditional weighted-growth enclosures and a regular-zero graph certificate with physical length normalization. Critical zeros and unbounded tilts remain genuine obstructions. Contribution assessment: C2/5, a modest source-specific calibration and higher-jet profile refinement, without a new global nodal-length theorem or a uniform nonlinear-gauge descent.

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1 The actual problem and its proof interfaces

Let (M, g) be a fixed connected closed C^∞ Riemannian surface and let u be a nonzero real eigenfunction

$$-\Delta_g u = \lambda u, \qquad \lambda > 0, \qquad k = \sqrt{\lambda}.$$

The assigned preprint [1] asserts $\mathcal{H}^1_g(\{u = 0\}) \le C_g k$, uniformly over all positive eigenspaces of this fixed metric. Orientability is not a hypothesis. Reading the entire assigned text is not an independent verification of this global assertion. The local results proved below neither supply the source's compactness and persistence arguments nor replace its external nodal theorem.

In an isothermal chart, $g = p(x)(dx_1^2 + dx_2^2)$, with $p > 0$ smooth, and $\Delta u + k^2 p u = 0$. A square Q of side s and center x_Q has wavelength parameter $K_Q = ks$. The source fixes $R = 1000$ and uses the *unnormalized Euclidean* L^2 norms

$$E(Q, b) = \log \frac{\|e^{-kb\cdot(x-x_Q)}u\|_{L^2(D(x_Q,Rs))}}{\|e^{-kb\cdot(x-x_Q)}u\|_{L^2(D(x_Q,s))}}, \quad m(Q, b) = \frac{E(Q, b)}{K_Q}, \quad N(Q) = m(Q, 0). \tag{1}$$

All disks must lie in the chart. The fixed spectral problem, unique continuation, and $u \neq 0$ ensure nonzero denominators in the source setting. Normalizing the disk norms by their areas would change E by $\log R$; we do not do so. The elementary untilting estimate is $N(Q) \le m(Q, b) + (R + 1)|b|$.

1.1 Architecture and ordered choices

The following is a map of the read proof, not a list of results newly established here.

1. `profiles.tex`, lines 1–452, constructs upper-semicontinuous logarithmic profiles from probability measures, with compact upper and open lower bounds. It proves subharmonicity for tilted eigenfunctions by a commutator argument, including tilts larger than the logarithmic scale. It then obtains fixed-open-set growth bounds.
2. `subdivision.tex`, lines 1–371, represents a normalized subharmonic profile by a harmonic function plus a logarithmic potential. Its affine correction gives mean profile excess $O(\rho^2(1 + |\log \rho|))$ and bounded mean slope. One first fixes the subdivision factor $A = \rho^{-1}$ and then obtains compactness thresholds.
3. `descent.tex`, lines 1–327, adds the floor $C_0/K_Q + s_Q$ to the tilted growth budget and stops on specified violations. The sum of floors along paths is controlled only while $K_Q \ge a_0 > 0$. Incoming tilts at stopping cells are retained.
4. `persistence.tex`, lines 1–532, converts large untilted excursions into witnesses, possibly outside the generating square. A $W^{1,1}$ affine approximation at a nonzero-gradient Lebesgue point uses subharmonicity as well as Poincare's inequality. A collar mismatch argument packs witnesses and gives bounded average terminal growth. It is not ordinary packing of subsets of their own squares.
5. `length.tex`, lines 1–138, rescales terminal cells to a small-potential Schrödinger problem, imports a nodal-length/growth theorem, and transfers Euclidean length back to g.

The introduction, driver, preamble, bibliography, and README were also read in full. The full source has no asserted effective computational threshold in the compactness step. Our profile estimates do not create one.

1.2 Large tilts are a real uniformity requirement

In the source's rescaled equation $\Delta U_j + K_j^2 p_j U_j = 0$, let $S_j \to \infty$, $B_j \in \mathbb{R}^2$, and $D_j = S_j + |B_j|$. The subharmonic-profile argument requires

$$\frac{K_j}{D_j} \longrightarrow 0, \qquad \frac{K_j^2 \|\nabla p_j\|_\infty}{S_j D_j} \longrightarrow 0, \qquad \sup_j \|p_j\|_\infty < \infty. \tag{2}$$

There is no imposed bound on $|B_j|/S_j$ or $\log D_j/S_j$. For a test weight $T = -B_j \cdot X - S_j \chi(X)$ the source splits the conjugated operator into its symmetric and skew parts. Its exact commutator includes

$$4T''(\nabla v, \nabla v) + (4T''(\nabla T, \nabla T) - \Delta^2 T + 2K_j^2 \nabla p_j \cdot \nabla T)v^2.$$

The cutoff estimates cancel the large prefactor before the exponential comparison. Replacing this with a crude absolute Taylor error multiplied by $e^{R|B_j|}$ would lose the very uniformity the source needs.

2 An exact calibration of the affine loss

Identify $\mathbb{R}^2$ with $\mathbb{C}$. Fix $R > 1$, $0 < \rho < 1$, and a center y. For a locally integrable subharmonic function V and a real polynomial P with $P(0) = 0$, put

$$\mathcal{E}_y(V, P) = \sup_{\overline{D(y,R\rho)}} \{V(x) - P(x - y)\} - \sup_{D(y,\rho)} \{V(x) - P(x - y)\}. \tag{3}$$

For the profiles considered below these suprema are finite; point values $-\infty$ at atoms cause no difficulty. This is a difference of suprema, not the full oscillation or an L^2 growth exponent.

Proposition 2.1 (One atom and the prescribed gradient)**.** *Let $V(x) = a \log|x - z| + c$, with $a > 0$ and $r = |y - z| > R\rho$. For $P(w) = \nabla V(y) \cdot w$,*

$$\mathcal{E}_y(V, P) = \frac{a(R^2 - 1)\rho^2}{2r^2}. \tag{4}$$

Proof. After rotating, the centered remainder on a circle of radius $s < r$ is $aH_q(\cos\theta)$, where $q = s/r$ and

$$H_q(v) = \tfrac{1}{2}\log(1 + 2qv + q^2) - qv, \qquad -1 \le v \le 1.$$

This function is strictly concave in v when $q > 0$; its derivative vanishes at $v = -q/2$, which belongs to the interval. Its maximum is $q^2/2$. The centered remainder is harmonic on the closed disk, so the same maximum holds over that disk. Supremum over the open inner disk equals the boundary maximum by continuity. Subtract the two values and restore the factor a. □

Corollary 2.2 (The logarithm is present for this rule)**.** *Let $R > 1$ be fixed, let A tend to infinity through odd integers, $\rho = A^{-1}$, and use the A^2 centers of the uniform subdivision of $[-1/2, 1/2]^2$. The normalized source-admissible profile*

$$V(x) = \frac{\log(|x|/R)}{\log R}, \qquad x \in D(0, R),$$

satisfies $V \le 0$ and $\sup_{\overline{D}(0,1)} V = -1$. On centers farther than $2R\rho$ from the atom, the source's affine correction is exactly its gradient. The average of their gaps is bounded below by $c_R\rho^2 \log A$ for all sufficiently large A.

Proof. Write a center as $y = \rho(j, \ell)$. For each integer $n_0 = \lfloor 2R \rfloor + 1 \le n \le \lfloor A/4 \rfloor$, the square ring $\max(|j|, |\ell|) = n$ has $8n$ centers. They satisfy $|y| > 2R\rho$ and $|y|^2 \le 2n^2\rho^2$. By (4), each gap is at least $(R^2 - 1)/(4n^2 \log R)$. Division by A^2 gives the lower bound

$$\frac{2(R^2-1)}{\log R}\rho^2 \sum_{n=n_0}^{\lfloor A/4 \rfloor} \frac{1}{n}.$$

The harmonic sum is $\log A + O_R(1)$. □

This does not optimize over all affine polynomials; it calibrates the particular gradient rule used in the source. Nor is this singular profile asserted to be realized by a sequence of eigenfunctions. It shows that discarding the logarithm from that profile estimate without changing the correction is not justified.

3 Harmonic jets with the singular part retained

Work in the source's interior normalization

$$V(x) = h(x) + \int_{D(0,4)} \log|x - z| \, d\nu(z), \qquad x \in D(0,2), \tag{5}$$

where ν is a finite positive measure, $\nu(D(0,4)) \le M$, and h is real harmonic on $D(0,3)$. Assume a bound on its L^1 norm there. On a smaller fixed disk this bounds every fixed derivative of h, by the interior harmonic estimates. In the source these bounds follow from the profile normalization and its Riesz decomposition. Positivity of ν, not an arbitrary signed moment list, is used below.

Take $R\rho < 1/10$ and the centers of the unit-square grid. For an integer $m \ge 1$, split sources at distance $2R\rho$ and define

$$M_j(y) = \int_{|z-y|>2R\rho} (y - z)^{-j} \, d\nu(z), \quad 1 \le j \le m, \tag{6}$$

$$P_{m,y}(w) = \sum_{j=1}^{m} H_{j,y}(w) + \operatorname{Re} \sum_{j=1}^{m} \frac{(-1)^{j+1}}{j} M_j(y) w^j. \tag{7}$$

Here $H_{j,y}$ is the homogeneous degree-j part of the real Taylor polynomial of $h(y + w)$. Each part is harmonic: the homogeneous components of the convergent harmonic expansion are harmonic. For each fixed ρ the far moments are absolutely integrable. We do not differentiate the cutoff in y, and we make no assertion that $y \mapsto P_{m,y}$ is smooth across a cutoff.

Theorem 3.1 (Higher-jet profile bound and its costs)**.** *For fixed $m \ge 1$ and $R > 1$, the actual pointwise representation (5) gives*

$$0 \le \mathcal{E}_y(V, P_{m,y}) \le C_{m,R} \left[\rho^{m+1} + \int \min\{1, (\rho/|z - y|)^{m+1}\} \, d\nu(z) \right]. \tag{8}$$

At $z = y$ the minimum is defined to be 1. Constants depend on the stated harmonic bound and M, but not on ρ, the atom locations, or an eigenspace. Averaging over the A^2 grid centers, $\rho = 1/A$, gives

$$\operatorname*{av}_y \mathcal{E}_y(V, P_{m,y}) \le \begin{cases} C_R\rho^2(1 + |\log\rho|), & m = 1, \\ C_{m,R}\rho^2, & m \ge 2. \end{cases} \tag{9}$$

The coefficient costs, in the usual Euclidean tensor norms with fixed degree-dependent constants, satisfy

$$\operatorname*{av}_{y} |D^j P_{m,y}(0)| \le C_{m,R} \begin{cases} 1, & j = 1, \\ 1 + |\log \rho|, & j = 2, \\ 1 + \rho^{2-j}, & 3 \le j \le m. \end{cases} \tag{10}$$

Proof. The lower bound follows from inclusion of the disks. The upper bound can be proved with the inner *disk average* in place of the inner supremum, which only increases it. Harmonic polynomials $P_{m,y}$ have disk average zero because $P_{m,y}(0) = 0$.

For a far source put $d = y - z$ and $w = x - y$. Since $|w/d| < 1/2$ on the outer disk, the convergent logarithm expansion has remainder

$$\left| \log|d + w| - \log|d| - \operatorname{Re} \sum_{j=1}^{m} \frac{(-1)^{j+1}}{j} (w/d)^j \right| \le \frac{(|w|/|d|)^{m+1}}{(m+1)(1 - |w|/|d|)}. \tag{11}$$

It is harmonic throughout that disk and has inner average zero. Thus its outer supremum is at most $2(R\rho/|d|)^{m+1}/(m+1)$. The harmonic remainder of h has the same zero-average property and an outer bound $C_{m,R}\rho^{m+1}$. The necessary derivative bound follows, for example, by representing h by its Poisson integral on an intermediate disk and differentiating the kernel on a strictly smaller disk; one first selects such a boundary with bounded integral by radial integration of the L^1 norm.

For a near source, $|d| \le 2R\rho$, do not expand the logarithm. Scaling $w = \rho\xi$ shows that its inner average is at least $\log \rho - 1/2$, since

$$\operatorname*{av}_{D(0,1)} \log|\xi - t| \, d\xi = \begin{cases} (|t|^2 - 1)/2, & |t| \le 1, \\ \log|t|, & |t| \ge 1. \end{cases} \tag{12}$$

Its outer supremum is at most $\log \rho + \log(3R)$. Consequently its contribution is at most $\log(3R) + 1/2$. The formula (12) follows by circular averaging of the logarithm, which gives $\log \max(s, |t|)$ on a circle of radius s, followed by $2\int_0^1 s \, ds$. Logarithmic singularities are area-integrable. Tonelli applied to an integrable lower shift, and then linearity, justifies the inner averages with arbitrary finite positive ν. Combining the near and far bounds proves (8); on the near region the minimum is at least $(2R)^{-(m+1)}$.

For completeness, uniformly in z the grid satisfies

$$\#\{y : |y - z| \le t\} \le C(1 + t^2/\rho^2).$$

The centers within $O(\rho)$ of z contribute $O(\rho^2)$ after division by A^2. On annuli $|y - z| \asymp 2^\ell \rho$, the contribution is at most $C\rho^2 2^{(1-m)\ell}$. There are $O(1 + |\log \rho|)$ nonempty relevant scales in the fixed bounded domain. For $m = 1$ this gives a logarithm; for $m \ge 2$ the sum is geometric. Integrate in z against ν to obtain (9).

Finally the absolute degree-j coefficient is bounded by a constant times $1 + \int_{|z-y|>2R\rho} |y - z|^{-j} \, d\nu(z)$. The same annular count now gives an average bounded by $1 + \rho^{2-j} \sum_{\ell \le O(\log(1/\rho))} 2^{(2-j)\ell}$. This yields exactly the three regimes in (10). □

Remark 3.2 (What is and is not improved). The $m = 1$ argument is the source's logarithmic-potential mechanism, not a new Ram theorem. The higher-degree argument uses classical harmonic Taylor expansion, the mean-value property, and planar grid counting. The local increment is an explicit comparison of the exact source correction, its logarithmic loss, and the cost of removing that loss. Already $m = 2$ removes the logarithm in the excess; increasing m does not improve the order ρ^2 in this estimate when arbitrary near atoms are allowed.

3.1 Finite moments and certified discarded sources

Proposition 3.3 (A separated-source budget)**.** *Suppose some far sources have been represented exactly, while an omitted positive measure has mass at most b and distance at least $L > 2R\rho$ from y. Its degree-j moment has absolute value at most bL^{-j}. If those moments are enclosed and included in the correction, its remaining contribution to the outer-minus-inner-average certificate is at most*

$$\frac{b(R\rho/L)^{m+1}}{(m+1)(1-R\rho/L)}. \tag{13}$$

If an approximate correction differs from $P_{m,y}$ by a polynomial whose supremum on $D(0,R\rho)$ is at most ϵ_P, its actual gap differs from $\mathcal{E}_y(V,P_{m,y})$ by at most $2\epsilon_P$. For moment errors $|\widehat{M}_j - M_j| \le e_j$ one may take $\epsilon_P = \sum_{j=1}^{m} e_j(R\rho)^j/j$, plus a separate harmonic Taylor coefficient error.

Proof. The moment estimate is the integral triangle inequality. Integrating (11) and using the separation gives (13). For the last statement, each of the two suprema changes by at most ϵ_P, including on the open inner disk; subtract and use the triangle inequality. □

This is a certificate only when the *whole* discarded measure has the specified mass and separation. Finite moment agreement alone does not bound missing near atoms. The original near measure must remain in the logarithmic potential or receive its full near-source budget.

4 Why the new correction is not a new descent

On a rescaled cell let U solve $\Delta U + K^2 pU = 0$ and set $v = e^{-SP}U$, for a real smooth P. Direct differentiation gives

$$\Delta v + 2S\nabla P\cdot\nabla v + (S\Delta P + S^2|\nabla P|^2 + K^2 p)v = 0. \tag{14}$$

For harmonic P the ΔP term vanishes, but the other new coefficients are not constant when $\deg P \ge 2$. In particular the drift has derivative $2SP''$ and the zeroth-order term contains $S^2|\nabla P|^2$. Formula (10) permits a logarithmic mean Hessian cost, not the source's bounded mean vector increment.

At one fixed finite subdivision, the source's weak profile bounds can accommodate a fixed continuous weight. That fact alone gives no uniform subharmonicity theorem for iteratively changing polynomial gauges. The source's descendants carry vectors b, and its stopped proof controls $\sum|\Delta b|$, not a collection of higher tensors. The persistence proof further converts a nonzero affine slope into large untilted growth. All of these interfaces would need new estimates before using Theorem 3.1 to change the nodal-length proof. In particular, the degree-m construction does not show that the source subdivision factor, thresholds, or global constant improve.

There is a useful scale distinction: a degree-j coefficient cost ρ^{2-j} becomes $O(\rho^2)$ when evaluated at $|w| = O(\rho)$. Small values on a child cell do not mean small unscaled derivatives or a bounded accumulated gauge along a path. This is precisely why the value remainder and coefficient budgets have been stated separately.

5 Original Ram jets and conditional growth enclosures

Two original private texts were freshly read locally: TR208 [4] and the original differentiated-moment chapter [5]. Their relevant content is exact Leibniz accounting and Taylor remainders, not a claim

that every finite derivative table determines a solution. This distinction is especially important for nodal geometry.

For $U(X) = u(x_0 + sX)$, write $p_s(X) = p(x_0 + sX)$ and $K = ks$. For any multi-index γ the exact differentiated equation is

$$\Delta D^\gamma U + K^2 \sum_{\delta \le \gamma} \binom{\gamma}{\delta} D^\delta p_s \, D^{\gamma-\delta} U = 0, \qquad D^\delta p_s = s^{|\delta|} (D^\delta p)(x_0 + sX). \tag{15}$$

It uses field derivatives two orders higher than γ. A finite jet system becomes square only after specified closure information; variable coefficients do not automatically make it triangular. For a C^{N+1} function on every segment from 0 to X, the directional Taylor formula gives

$$|U(X) - T_N U(X)| \le \frac{\|X\|_1^{N+1}}{(N+1)!} \max_{\substack{|\alpha|=N+1 \\ 0 \le t \le 1}} |D^\alpha U(tX)|. \tag{16}$$

This follows by applying one-dimensional Taylor's formula to $U(tX)$ and the multinomial theorem. Applied to $D_i U$, it gives a derivative error but requires the corresponding extra derivatives.

In the original moment chapter, differentiating a fixed-window row also differentiates the kernel and the field remainder. If an operator uses derivatives up to order p, the omitted field orders and the needed recovery regularity must be retained. A polynomial recovery matrix invertible on $\mathcal{P}_N$ controls the component in that class, not an unrestricted smooth function. TR208's positive elliptic energy theorem is for a coercive boundary problem; it is not a ready-made inverse for $\Delta + K^2 p$ near a spectral eigenvalue. These are preserved valid restrictions of the original proofs.

Proposition 5.1 (An actual weighted norm enclosure)**.** *Let $U, P \in L^2(D(0,R))$, let $B \in \mathbb{R}^2$ be fixed, and suppose*

$$\|e^{-B\cdot X}(U - P)\|_{L^2(D(0,r))} \le \epsilon_r, \qquad r = 1, R.$$

Put $a_r = \|e^{-B\cdot X} P\|_{L^2(D(0,r))}$. If $a_1 > \epsilon_1$, then the growth in these two disks is well defined and

$$E \le \log \frac{a_R + \epsilon_R}{a_1 - \epsilon_1}. \tag{17}$$

If also $a_R > \epsilon_R$, then $E \ge \max\{0, \log((a_R - \epsilon_R)/(a_1 + \epsilon_1))\}$. These statements apply to the source's $E(Q, b)$ after rescaling, with $B = Kb$; the factor s in both L^2 norms cancels.

Proof. Use the triangle and reverse triangle inequalities in each *weighted* physical disk norm. The positive lower denominator also proves that U is not zero there. Divide the valid bounds, use the monotonicity of the logarithm, and use inclusion of disks for $E \ge 0$. □

No orthogonality of polynomial coefficients was assumed. The numbers a_r involve the actual exponential weight and disk Gram matrix. A uniform pointwise error ϵ gives only $\epsilon_r \le \epsilon e^{r|B|}\sqrt{\pi}\, r$ by a crude bound. That can be useless in (2). One needs an actual weighted residual or additional derivative/tilt information. A finite low-order Gram calculation cannot provide that information for all omitted derivatives or all tilts.

6 A certificate on regular nodal cells

The next statement is deliberately conditional and local. Unlike a bare jet comparison, it gives an actual zero-set conclusion, because it contains a uniform transversality assumption and boundary signs.

Theorem 6.1 (Regular graph and physical length)**.** *Let $\mathcal{R} = [-a, a] \times [-b, b]$, $a, b > 0$, and let $U, T \in C^1$ on a neighborhood of $\mathcal{R}$. Suppose*

$$\|U - T\|_\infty \le \epsilon_0, \qquad \|\partial_i(U - T)\|_\infty \le \epsilon_1 \quad (i = 1, 2), \qquad \partial_2 T \ge g > \epsilon_1$$

throughout $\mathcal{R}$, and, for every $x \in [-a, a]$,

$$T(x, -b) < -\epsilon_0, \qquad T(x, b) > \epsilon_0.$$

Then all zeros of U in $\mathcal{R}$ form a single C^1 graph $y = \phi(x)$, and, with $L = \|\partial_1 T\|_\infty$,

$$\mathcal{H}^1(\{U = 0\} \cap \mathcal{R}) \le 2a\sqrt{1 + \left(\frac{L + \epsilon_1}{g - \epsilon_1}\right)^2}. \tag{18}$$

The zero graph ϕ_T of T also exists, and $\|\phi - \phi_T\|_\infty \le \epsilon_0/g$. In a physical chart $x_{\text{phys}} = x_0 + sX$, $g_{\text{phys}} = p(x_{\text{phys}})|dx_{\text{phys}}|^2$, the right side of (18) is multiplied by $s\sqrt{\sup_{\mathcal{R}} p(x_0 + sX)}$.

Proof. The hypotheses imply opposite strict signs for U on the horizontal edges and $\partial_2 U \ge g - \epsilon_1 > 0$. The intermediate value theorem and strict monotonicity give exactly one zero on each vertical segment. The implicit function theorem, including local extensions at the two end values of x, gives $\phi' = -\partial_1 U/\partial_2 U$. This proves the slope bound and the arc-length estimate. Apply the same argument to T. At the actual zero of U, $|T| \le \epsilon_0$; integrate $\partial_2 T \ge g$ between the two zero heights to obtain the position bound. Finally the physical line element is $s\sqrt{p(x_0 + sX)}\,|dX|$, giving the stated metric factor. □

Example 6.2 (A cell that is genuinely certifiable)**.** *On $[-1/2, 1/2] \times [-1, 1]$ take $T(x, y) = y - x^2/4$ and $U(x, y) = T(x, y) + \eta \sin x$, $|\eta| \le 1/10$. Then $\epsilon_0 = \epsilon_1 = 1/10$, $g = 1$, and $L = 1/4$ are valid bounds. Both boundary signs are strict, the graph is $\phi(x) = x^2/4 - \eta \sin x$, and the certified Euclidean length is at most $\sqrt{1 + (7/18)^2}$. This manufactured smooth cell is not claimed to solve the eigenfunction equation. Its role is to test the full boundary and transversality certificate, not merely a matrix of Taylor coefficients.*

6.1 Negative controls and the critical set

For any finite jet order N, the harmonic polynomial $U_m(x, y) = \mathrm{Re}(x + iy)^m$, $m > N$, has zero jet of order N at the origin but $2m$ nodal rays. Its nodal length in a disk of radius r is $2mr$, and its supremum on the unit disk is 1. Thus even an exact elliptic equation and a finite center jet do not by themselves bound nodal length. Here the frequency parameter of the equation is zero, so this is not a counterexample to the positive-eigenvalue theorem on a closed surface. The missing information is growth and regularity of the zero set at the critical point, not an inaccurate polynomial evaluation.

Nor is an absolute small residual sufficient. On a fixed rectangle, let $U_j(x, y) = j^{-r-2} \sin(jx)$, where $r \ge 2$ is fixed. Then $\|U_j\|_{C^r} = O(j^{-2})$ and $\|(\Delta + K^2 p)U_j\|_\infty \to 0$ for fixed bounded p and K. Nevertheless the number of vertical nodal segments grows linearly with j. These are not exact eigenfunctions; they show why a small *absolute* equation residual and small finite jets cannot replace the missing normalization and nodal transversality. At critical zeros Theorem 6.1 must simply be marked inapplicable. Summing only certified regular cells leaves an uncontrolled critical-cell complement unless a separate argument handles it.

7 The unchanged passage to the global claim

The assigned proof obtains a uniform terminal average of N using its affine stopped process and witness packing. It does not obtain that average from a finite polynomial approximation. To see the exact role of this input, let a root square have side s_0 and terminal side s with $a_0 \le ks \le \varepsilon$, where $a_0 > 0$ is fixed only after the subdivision factor and the small-potential scale have been chosen. There are $(s_0/s)^2$ terminal cells, and the source's imported terminal-average result gives $\operatorname{av} E(I,0) \le Cks$.

The external local ingredient is Roy-Fortin's Theorem 2.1.1 [2]. It bounds the nodal length in $D_{1/60}$ of a real solution of $\Delta F + qF = 0$ on D_3 by $C\max(1, \log(\sup_{D_{5/2}} |F| / \sup_{D_{1/4}} |F|))$, when q is smooth and sufficiently small in supremum norm. The sign of q is unrestricted. Its full proof is imported. The source uses $F(z) = u(x_I + 60sz)$, so $q(z) = (60ks)^2 p(x_I + 60sz)$. Its disks of radii $180s$, $165s$, and $15s$ fit inside the available Rs disk. Interior elliptic estimates convert the supremum growth to $C + E(I,0)$. Therefore the Euclidean length in I is at most $Cs(1 + E(I,0))$. Summing gives

$$\sum_I Cs(1 + E(I,0)) \le Cs_0^2(1/s + k) \le Cs_0^2(a_0^{-1} + 1)k. \tag{19}$$

The finite atlas and the physical metric factor complete the *source's* asserted conclusion. The estimates of Sections 2–6 do not independently prove the terminal-average premise of (19).

8 Comparison, costs, and future work

The improvement of Theorem 3.1 is exact at a specified interface: far logarithmic sources can be corrected to higher harmonic order while singular near sources remain intact. Its advantages are an $O(\rho^2)$ mean remainder, explicit moment coefficients, an exact atomic calibration, and a separate discarded-mass budget. These features can help check a proposed finite subdivision or distinguish value accuracy from derivative accuracy.

The costs are equally specific. Positive Riesz mass and a harmonic interior bound are required. Coefficients depend on the cell and on a cutoff. Higher derivatives become more expensive, and the nonlinear gauge in (14) is not the source's affine operator. Near atoms can still contribute an order-ρ^2 mean error. No experiment here establishes a smaller admissible A, a faster algorithm, a quantitative compactness threshold, or a new historical result about Yau's conjecture.

An admissible next step would be a polynomial-gauge subharmonicity theorem with uniform control of the new drift derivatives, followed by a stopped descent carrying all required tensors. It would have to handle changing gauges and their accumulated costs, rather than prove a one-cell identity. Another direction is to combine Theorem 6.1 with a genuine critical-cell estimate and source-normalized weighted growth bounds. Neither missing result is assumed here. Smoothness of the fixed metric gives local derivative bounds, but it is not a quantitative bound on all eigenspace jets at wavelength scale.

8.1 Primary literature and status

The publisher's README freshly accessed on October 9, 2026 identifies the assigned work as an OpenAI Math Release preprint dated September 23, 2026. That is publication metadata, not peer-review or independent proof certification. Roy-Fortin's published 2015 theorem was checked at its actual statement, including disk radii and small-potential hypotheses; its full external proof was not audited. The 2015 L^q companion note [3] was inspected at abstract and version history only: its stated extension concerns average local growth for $1 < q < \infty$. It is context, not a substitute

for the assigned source's terminal-growth estimate. No exhaustive priority search or independent acceptance claim is made.

9 Reproducibility and evidence boundaries

The author literally read all nine public TeX files and the README, 2175 lines in total, in bounded untruncated displays. Freshly read private original intervals were TR208 lines 143–380 and 543–594, and the differentiated-jet chapter lines 26–160, 513–676, and 747–941: 784 lines in total. Their selected-file unread complements contain 2120 lines. Other private files and included dependencies were not read merely because a citation or index mentioned them. The ledger preserves pre-read hashes, successful display intervals, post-read hashes, and exact complements; no whole-corpus read is claimed.

Owned executable checks target the atomic derivative identity, harmonicity and coefficient signs of logarithmic jets, the exact conjugation formula, square-ring counting, separated-tail sums, and transversality fixtures with negative controls. High-precision numerical diagnostics are distinguished from exact arithmetic. The executed owned suite passed 39 exact cases with 147 assertions and 16 numerical cases with 27 assertions at 70 decimal digits. They do not certify any profile realization, compactness theorem, uniform persistence estimate, or global nodal theorem. Parent review and parent fixtures are separate evidence and are not included in the author's check totals.

References

[1] OpenAI, *Sharp nodal length on smooth surfaces*, Math Release preprint, September 23, 2026. Publisher record. Assigned cached source: `profiles`, `subdivision`, `descent`, `persistence`, and `length`, with driver, introduction, preamble, bibliography, and README.

[2] G. Roy-Fortin, *Nodal sets and growth exponents of Laplace eigenfunctions on surfaces*, Analysis & PDE **8** (2015), 223–255. Primary published PDF, Theorem 2.1.1. Statement and adjacent normalization paragraphs inspected.

[3] G. Roy-Fortin, *Average L^q growth and nodal sets of eigenfunctions of the Laplacian on surfaces*, arXiv:1510.02376v1, October 8, 2015. Abstract and version metadata only.

[4] M. Subbarao, original confidential TR208, *Multidimensional variable-coefficient operators*, `TR208-multidimensional-variable-coefficient-operators.tex`, lines 143–380 and 543–594 freshly read. Exact multi-index Leibniz rows, directional Taylor remainder, and conditional elliptic energy inversion. Local original, not publicly transmitted.

[5] M. Subbarao, original confidential nonlinear integral/PDE report, revised TR1, `sec-03-jets.tex`, lines 26–160, 513–676, and 747–941 freshly read. Moment expansions, differentiated remainders, moving boundaries, and finite polynomial recovery with omitted-order control. Local original, not a later comparative summary.